

Tuesday
09/12/25

(1.) Consider a table with arrival time and burst time:-

Process	Arrival Time	Burst Time
P_1	0	5
P_2	0	3
P_3	0	8

(a.) Find the average turn around time and waiting time?

Ans) Turnaround time = Completion time - Arrival time
waiting time = Turn around time - Burst Time

$$\rightarrow \text{TAT of } P_1 = 5$$

$$\text{TAT of } P_2 = 8$$

$$\text{TAT of } P_3 = 16$$

$$\text{Average TAT} = \frac{5+8+16}{3} = \frac{29}{3} = \underline{\underline{9.67}}$$

$$\rightarrow \text{W.T of } P_1 = 0$$

$$\text{W.T of } P_2 = 5$$

$$W.T \text{ of } P_3 = 8$$

$$\text{Average } W.T = \frac{0+5+8}{3} = \frac{13}{3} = \underline{\underline{4.3}}$$

(2) Consider the table:-

P ₁	P ₂	P ₃
0	5	8
		16

Process	Burst Time	Arrival Time
P ₁	6	0
P ₂	8	0
P ₃	3	0

Use SJF and find Average Turnaround time and waiting Time?

Ans.) P₃ = 3 is the shortest

P ₃	P ₁	P ₂
0	3	9
		17

TAT = Completion time - Arrival time

$$TAT \text{ of } P_3 = 3 - 0 = \underline{\underline{3}}$$

$$TAT \text{ of } P_1 = 9 - 0 = \underline{\underline{9}}$$

$$TAT \text{ of } P_2 = 17 - 0 = \underline{\underline{17}}$$

→ Waiting Time = TAT - Burst Time

$$WT \text{ of } P_3 = 3 - 3 = \underline{\underline{0}}$$

$$WT \text{ of } P_1 = 9 - 6 = \underline{\underline{3}}$$

$$WT \text{ of } P_2 = 17 - 8 = \underline{\underline{9}}$$

$$\text{Average TAT} = \frac{3 + 9 + 17}{3} = \frac{29}{3} = \underline{\underline{9.67}}$$

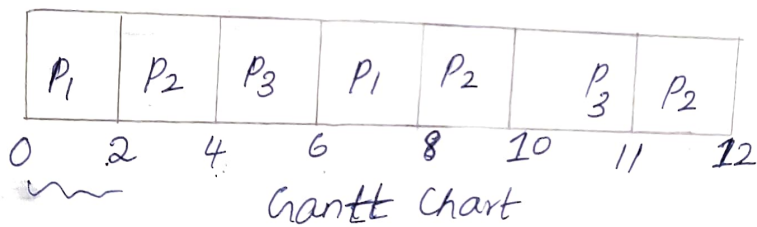
$$\text{Average WT} = \frac{0 + 3 + 9}{3} = \frac{12}{3} = \underline{\underline{4}}$$

15/12/25
Monday (.) Use Round Robin and find the Average Turn around time and waiting time.

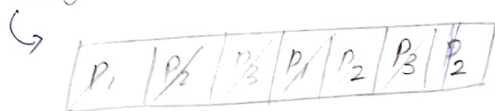
Process	Burst Time	Arrival Time
P_1	4	0
P_2	5	0
P_3	3	0

Time Quantum = 2 [switching time]

Ans.)



Ready Queue



P_1

Turnaround Time = 8

waiting Time = 4

P₂

Turn around time = 12

waiting Time = 7

P₃

Turn around time = 11

waiting Time = 8

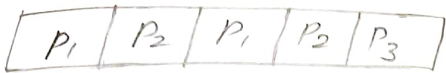
$$\begin{aligned}\text{Average Turn around Time} &= 11 + 12 + 8 \\ &= \frac{31}{3} = \underline{\underline{10.3}}\end{aligned}$$

$$\begin{aligned}\text{Average waiting Time} &= \frac{7 + 8 + 4}{3} \\ &= \frac{19}{3} = \underline{\underline{6.3}}\end{aligned}$$

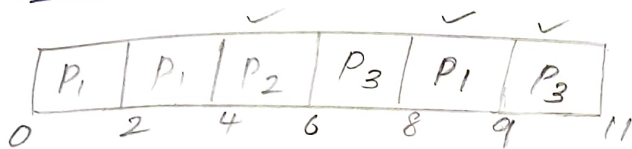
(2.) Use Round Robin, Time slice = 2

Process	Burst Time	Arrival Time
P ₁	5	0
P ₂	2	4
P ₃	4	5

Ans.)



Gantt chart



P_1

Turn around time = ~~14~~ 9

Waiting Time = $9 - 5 = \underline{\underline{4}}$

p_2

Turn around time = 2

waiting Time = 0

P_3

Turn around time = 6

waiting Time = 2

Average Turn Around time = 5.6

Average waiting Time = 2

(3.) Use Round Robin, Time slice = 4

Process	Arrival time	waiting Time
P_1	2	8
P_2	0	7
P_3	1	9

Ans. Ready Queue

P_2	P_3	P_1	P_2	P_3	P_1	P_3
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Gantt Chart

P_2	P_3	P_1	P_2	P_3	P_1	P_3	
0	4	8	12	15	19	23	24

P_1

$$TAT = 23 - 2 = \underline{\underline{21}}$$

$$WT = 21 - 8 = \underline{\underline{13}}$$

P₂

$$TAT = 15 - 0 = \underline{\underline{15}}$$

$$WT = 15 - 7 = \underline{\underline{8}}$$

P₃

$$TAT = 24 - 1 = \underline{\underline{23}}$$

$$WT = 23 - 9 = \underline{\underline{14}}$$

$$\text{Avg } TAT = \frac{21 + 15 + 23}{3}$$

$$\Rightarrow \frac{59}{3} = \underline{\underline{19.6}}$$

$$\begin{array}{r} 11. \\ 3 \overline{) 35} \\ \underline{33} \\ 25 \\ \underline{24} \\ 10 \end{array}$$

$$\text{Avg } WT = \frac{13 + 8 + 14}{3} = \frac{35}{3} = \underline{\underline{11.6}}$$

(4.) Use SJF scheduling to find avg TAT and WT :-

Process	Arrival time	Burst Time
P ₁	0	6
P ₂	4	10
P ₃	4	4
P ₄	8	3

Tuesday.

16/12/25

(1.) Use preemptive SJF / SRTF

Preemptive
(Interpt)

Process	Arrival Time	Burst Time
P ₁	0	8
P ₂	4	10
P ₃	4	3
P ₄	10	4

FCFS - purely non-preemptive

SJF - Non-preemptive

SCTF - preemptive ← preemptive.

RR - purely preemptive.

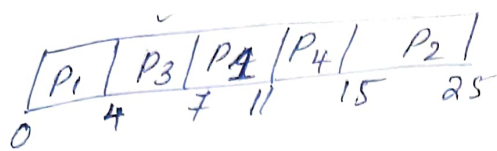
Priority - can be either preemptive or non-preemptive

Ans.)

Ready Queue

P ₁	P ₃	P ₂	P ₄
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Gantt chart



P

TAT

$$\text{TAT of } P_1 = 11 - 0 = \underline{\underline{11}}$$

$$\text{TAT of } P_2 = 25 - \overset{4}{\underset{\wedge}{11}} = \underline{\underline{14}}$$

$$\text{TAT of } P_3 = 7 - 4 = \underline{\underline{3}}$$

$$\text{TAT of } P_4 = 15 - 10 = \underline{\underline{5}}$$

$$\text{Average TAT} = \frac{40}{4} = \underline{\underline{10}}$$

W.T = TAT - Burst time

$$\text{WT of } P_1 = 11 - 8 = \underline{\underline{3}}$$

$$\text{WT of } P_2 = 25 - \overset{1}{\underset{\wedge}{11}} - 10 = \underline{\underline{14}}$$

$$\text{WT of } P_3 = 7 - 7 = \underline{\underline{0}}$$

$$\text{WT of } P_4 = \overset{5}{\underset{\wedge}{15}} - 4 = \underline{\underline{11}}$$

$$\text{Average WT} = \frac{28}{4} = 7$$

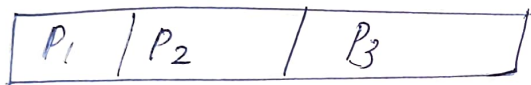
$$4 \overline{) 33} \begin{array}{r} 8 \\ 32 \\ \hline 1 \end{array}$$

* Priority Scheduling

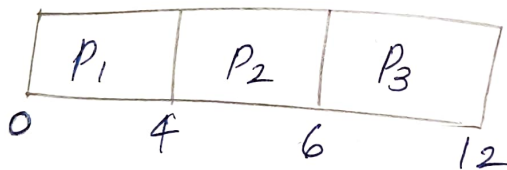
(2) Use priority Scheduling (non preemptive) and calculate avg TAT & WT

Process	Arrival Time	Burst Time	Priority
P ₁	0	4	2
P ₂	1	2	1
P ₃	2	6	3

Ans.) Ready Queue



Gantt Chart



$$P_1 \text{ TAT} = 4 - 0 = \underline{\underline{4}}$$

$$\text{TAT of } P_2 = 6 - \underset{1}{1} = \underline{\underline{5}}$$

$$\text{TAT of } P_3 = 12 - 2 = \underline{\underline{10}}$$

$$\text{Avg} = \frac{4 + 5 + 10}{3} = \frac{19}{3} = \underline{\underline{6.33}}$$

W.T

$$WT \text{ of } P_1 = 4 - 4 = 0$$

$$WT \text{ of } P_2 = 5 - 2 = 3$$

$$WT \text{ of } P_3 = 10 - 6 = 4$$

$$Avg = \frac{7}{3} = \underline{\underline{2.33}}$$

(3) Use Priority Scheduling (Preemptive) and find Avg TAT & Avg w.T?

Process	Arrival Time	Burst Time	Priority
P ₁	0	8	3
P ₂	4	10	2
P ₃	4	3	3
P ₄	10	4	1

Ans) Ready Queue

[P₁ | P₂ | P₄ | P₂ | P₁ | P₃]

P₁ - P₃ - same priority

Gantt Chart

[P₁ | P₂ | P₄ | P₂ | P₁ | P₃]
0 4 10 14 18 22 25

TAT

$$P_1 = 22 - 0 = \underline{\underline{22}}$$

$$P_2 = 18 - 4 = \underline{\underline{14}}$$

$$P_3 = 25 - \overset{4}{\cancel{22}} = \underline{\underline{21}}$$

$$P_4 = 14 - 10 = \underline{\underline{4}}$$

$$\text{Avg} = 22 + 14 + 21 + 4 = \frac{\cancel{22}}{4} = \underline{\underline{15.25}}$$

WT

$$P_1 = 22 - 8 = \underline{\underline{14}}$$

$$P_2 = 14 - 10 = \underline{\underline{4}}$$

$$P_3 = 21 - 3 = \underline{\underline{18}}$$

$$P_4 = 4 - 4 = \underline{\underline{0}}$$

$$\text{Avg} = \frac{14 + 4 + 18 + 0}{4} = \frac{36}{4} = \underline{\underline{9}}$$

(4.) Use priority (preemptive)

Process	Arrival Time	Burst Time	Priority
P ₁	0	6	2
P ₂	4	10	1
P ₃	4	4	2
P ₄	8	3	1

(5) Use preemptive / SJF / SRTF

Process	Arrival	Burst
P ₁	0	6
P ₂	1	3
P ₃	2	7

Ans. (4) Ready Queue

P ₁	P ₂	P ₄	P ₃	P ₁	P ₃
----------------	----------------	----------------	----------------	----------------	----------------

TA

Gantt Chart

P_1	P_2	P_4	P_1	P_3	
0	4	14	17	19	23

$$\underline{\text{Turn Around Time (TAT)}} = CT - AT$$

$$TAT \text{ of } P_1 = 19 - 0 = \underline{\underline{19}}$$

$$TAT \text{ of } P_2 = 14 - 4 = \underline{\underline{10}}$$

$$TAT \text{ of } P_3 = 23 - 4 = \underline{\underline{19}}$$

$$TAT \text{ of } P_4 = 17 - 8 = \underline{\underline{9}}$$

$$Avg = \frac{19 + 10 + 19 + 9}{4} = \frac{57}{4} = \underline{\underline{14.25}}$$

$$\text{waiting Time (WT)} = TAT - BT$$

$$WT \text{ of } P_1 = 19 - 6 = \underline{\underline{13}}$$

$$WT \text{ of } P_2 = 10 - 10 = 0$$

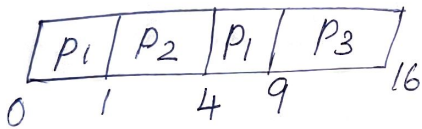
$$WT \text{ of } P_3 = 19 - 4 = \underline{\underline{15}}$$

$$WT \text{ of } P_4 = 9 - 3 = \underline{\underline{6}}$$

$$\text{Average WT} = \frac{13 + 15 + 6 + 0}{4} = \underline{\underline{8.5}}$$

(5.)

Process	Arrival time	Burst Time
P_1	0	6
P_2	1	3
P_3	2	7

Ready Queue $\boxed{P_1 \mid P_2 \mid P_3}$ Gantt ChartTAT

$$P_1 = 9 - 0 = \underline{\underline{9}}$$

$$P_2 = 4 - 1 = \underline{\underline{3}}$$

$$P_3 = 16 - 2 = \underline{\underline{14}}$$

$$\text{Avg} = \frac{9+3+14}{3} = \frac{16}{3} = \underline{\underline{8.67}}$$

WT

$$P_1 = 9 - 6 = \underline{\underline{3}}$$

$$P_2 = 3 - 3 = \underline{\underline{0}}$$

$$P_3 = 14 - 7 = \underline{\underline{7}}$$

$$\text{Avg} = \frac{3+0+7}{3} = \frac{10}{3} = \underline{\underline{3.33}}$$

★ Banker's Algorithm

used for deadlock Detection & Avoidance.

(Q1) Total -

A	B	C
10	5	7

Remaining next?

Process	Allocation ⁿ			Max Need			Available		
	A	B	C	A	B	C	A	B	C
P ₁	0	1	0	7	5	3	3	3	2 - unus
P ₂	2	0	0	3	2	2	5	3	2
P ₃	3	0	2	9	0	2	5	3	2 - unus
P ₄	2	1	1	4	2	2	7	4	3
P ₅	0	0	2	5	3	3	0	0	2
	7	2	5				7	4	5

$$745 +$$

$$755 +$$

$$302$$

$$1057 \quad A \quad B \quad C$$

	A	B	C	
Available $\Rightarrow$	10	5	7	-
	7	2	5	
	3	3	2	

Execution Order :

$\Rightarrow$ Safe order :

Unsafe.

$P_2 \rightarrow P_4 \rightarrow P_5 \rightarrow P_1 \rightarrow P_3$

(2) Check if the System is in safe state or in unsafe state using Banker's Algorithm.

Process	Allocation A B C	Max Need A B C	Available
P_0	1 1 2	4 3 3	$ \begin{array}{r} 2 \ 1 \ 0 + \\ 1 \ 1 \ 2 \text{ release} \\ \hline 3 \ 2 \ 2 + \\ 1 \ 1 \ 2 \\ \hline 4 \ 3 \ 4 + \\ 2 \ 1 \ 2 + \\ \hline 6 \ 4 \ 6 + \\ 4 \ 0 \ 1 + \\ \hline 10 \ 4 \ 7 + \\ 0 \ 2 \ 0 \\ \hline 10 \ 6 \ 7 \\ \hline \hline \end{array} $
P_1	2 1 2	3 2 2	
P_2	4 0 1	5 0 1	
P_3	0 2 0	7 3 3	
P_4	1 1 2	1 1 2	
	8 5 7		

Total Resource Count :-

A	B	C
10	6	7

$P_4 \rightarrow P_0 \rightarrow P_1 \rightarrow P_2 \rightarrow P_3$ - safe.

Process	Allocation matrix				max matrix				need matrix				Available
	A	B	C	D	A	B	C	D	A	B	C	D	
✓P ₀	0	1	1	0	0	2	1	0	0	1	0	0	1520 0110+
✓P ₁	1	2	3	1	1	6	5	2	0	4	2	1	16307 0632+
✓P ₂	1	3	6	5	2	3	6	6	1	0	0	1	11262 0014+
✓P ₃	0	6	3	2	0	6	5	2	0	0	2	0	11276 1231+
P ₄	0	0	1	4	0	6	5	6	0	6	4	2	214107 1265 3171612

2 12 14 12

max Resource count is

	A	B	C	D
	3	17	16	12

Create, need matrix = Max - Allocation. Also

Test the system is in safe state or not.

Ans) Available =

3	17	16	12	-
2	12	14	12	
1	5	2	0	

$P_0 \rightarrow P_3 \rightarrow P_4 \rightarrow P_1 \rightarrow P_2$

Thursday.
01/11/26

(1.) Consider the following snapshot of a system:

	<u>Allocation</u>	<u>Max</u>	<u>Available</u>
	A B C D	A B C D	A B C D
T_0	0 0 1 2	0 0 1 2	1 5 2 0
T_1	1 0 0 0	1 7 5 0	
T_2	1 3 5 4	2 3 5 0	
T_3	0 6 3 2	0 6 5 2	
T_4	0 0 1 4	0 6 5 6	

Answer the following questions using the Banker's Algorithm.

- (a.) what is the content of the matrix Need?
- (b.) Is the system in a safe state?
- (c.) If a request from thread T_1 arrives for $(0, 4, 2, 0)$, can the request be granted immediately?

Ans. (a.) Need matrix

Need = max - Allocation

	A	B	C	D
T_0	0	0	0	0
T_1	0	7	5	0
T_2	1	0	0	2
T_3	0	0	2	0
T_4	0	6	4	2

(b.) Yes, the system is in a safe state.

$T_0 \rightarrow T_3 \rightarrow T_2 \rightarrow T_4 \rightarrow T_1$

(c.) Request $\leq$ Need (T_1) ✓

Request $\neq$ Available.

Request cannot be granted immediately,
leads to unsafe state.

★ Page Replacement Algorithm

memory manager of OS select a page using page replacement Algorithms.

- If the page was modified or dirty, writing it back to disc before replacement.

→ page Replacement Techniques / Algorithms:-

(1) FIFO (first In First Out)

Consider page Reference string;

6, 1, 1, 2, 0, 3, 4, 6, 0, 2, 1, 2

with 3 page frames. Find the number of page fault using FIFO page Replacement Algorithm?

Miss - M, Hit - H

	6	1	1	2	0	3	4	6	0	2	1	2
f ₃				2	2	2	4	4	4	2	2	2
f ₂		1	1	1	1	3	3	3	0	0	0	0
f ₁	6	6	6	6	0	0	0	6	6	6	1	1
	M	M	H	M	M	M	M	M	M	M	M	H

Total ^{no. of} page fault (Miss) = 10

(2) Most recently used (MRU) Algorithm / Most frequently used.

Consider the page reference string

7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 3 with

four page frames. Find the number of page fault, page hit and Ratio of Page Hit to page

fault using MRU page Replacement

Algorithms.

	7	0	1	2	0	3	0	4	2	3	0	3	2	3
f_4				2	2	2	2	2	2	3	0	3	2	3
f_3			1	1	1	1	1	1	1	1	1	1	1	1
f_2		0	0	0	0	3	0	4	4	4	4	4	4	4
f_1	7	7	7	7	7	7	7	7	7	7	7	7	7	7
	M	M	M	M	H	M	M	M	H	M	M	M	M	M

Total no. of page fault = 12

Total no. of page Hit = 2

$$\text{Page Hit : page fault} = 2 : 12 = 1:6$$

[Hit Rate percentage] Page Hit percentage

$$\Rightarrow \frac{2}{14} \times 100 = \frac{100}{7}$$

$$\text{Page fault (Miss) percentage} = \frac{12}{14} \times 100$$

$$= \frac{600}{7}$$

(3.) Least Recently Used (LRU)

Consider the page reference string :-

7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 3 with 4 page frames. Find the page fault & page Hit percentage using LRU Replacement Algorithm?

	7	0	1	2	0	3	0	4	2	3	0	3	2	3
F ₄				2	2	2	2	2	2	2	2	2	2	2
F ₃			1	1	1	1	1	4	4	4	4	4	4	4
F ₂		0	0	0	0	0	0	0	0	0	0	0	0	0
F ₁	7	7	7	7	7	3	3	3	3	3	3	3	3	3
	M	M	M	M	H	M	H	M	H	H	H	H	H	H

Total no. of page hits = 8

Total no. of page fault = 6

$$\text{Page Hit Percentage} = \frac{8}{14} \times 100 = \frac{400}{7}$$

$$\text{Page Miss percentage} = \frac{6}{14} \times 100 = \frac{300}{7}$$

$$\text{Page Hit : page fault} = 8 : 6 = \underline{\underline{4 : 3}}$$

It breaks the expectation that more memory equals fewer number of faults.

Monday
16/02/26

★ Memory Allocation Strategies

(1) First fit

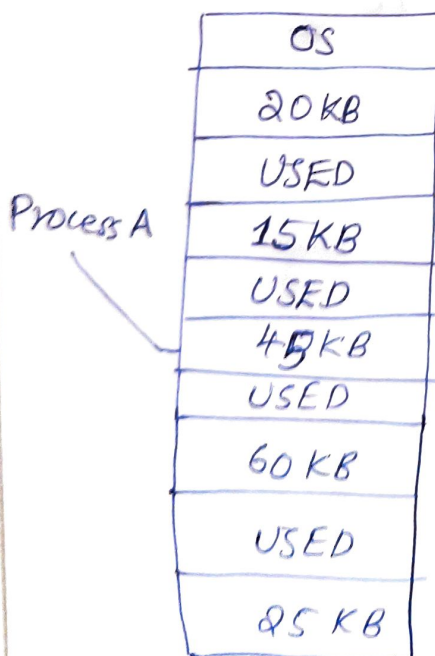
(2) Best fit

(3) Worst fit

(4) Next fit.

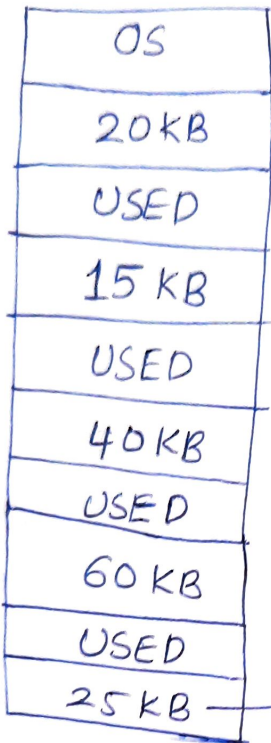
Internal
External Fragmentation

(1) First fit



$$\text{Hole} = 45 - 25 = \underline{\underline{15 \text{ KB}}}$$

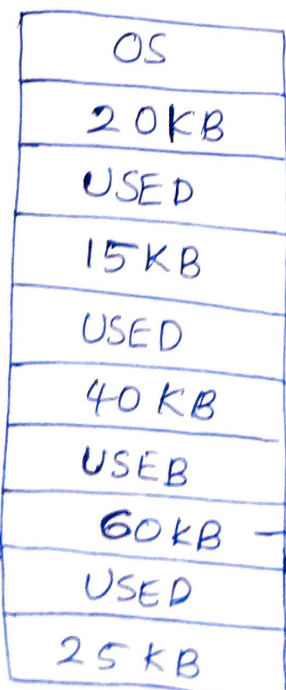
(★) Best fit



$$\text{Hole} = 25 - 25 = \underline{\underline{0 \text{ KB}}}$$

Process A

(★) WORST FIT



PROCESS A

• if not, then leave that and keep checking

⇒ Next fit is same as first fit.

(1) Consider 5 memory partitions of size,
100 KB, 500 KB, 200 KB, 450 KB and 600 KB

If sequence of Request are of size 212 KB,
417 KB, 112 KB and 426 KB in same order.

Then, Find the ^{m/m} allocation using first fit, Best
fit and worst fit?

Ans.) Given,

Block size = {100KB, 500KB, 200KB, 450KB, 600KB}

Process size = {212KB, 417KB, 112KB, 426KB}

First fit

Process No.	Process Size	Block No	Mlm wastage
1	212	2 (500)	$500 - 212 = 288$
2	417	4 (450)	$450 - 417 =$
3	112	3 (200)	$200 - 112 =$
4	426	5 (600)	$600 - 426 =$

Best fit

Process No	Process Size	Block No	Waste/memory wastage
1	212	4	
2	417	2	
3	112	3	
4	426	5	

Worst fit

Process No	Process Size	Block No	memory wastage
1	212	5	
2	417	2	
3	112	4	
4	426	Unallocated.	

* Disc scheduling Algorithms

(1) FCFS (first Come first served)

Consider the disc head moves for the sequence of request : 98, 183, 37, 122, 14, 124, 65, 67 and initial position of head is 53.

Ans) Initial head position = 53

~~53 98~~

Request sequence : 98, 183, 37, 122, 14, 124, 65, 67

$$\begin{aligned} \text{Total head movement} &\Rightarrow |53 - 98| + |98 - 183| + \\ &|183 - 37| + |37 - 122| + |122 - 14| + |14 - 124| + \\ &|124 - 65| + |65 - 67| \end{aligned}$$

$$45 + 85 + 146 + 85 + 108 + 110 + 59 + 2$$

$$\Rightarrow \underline{\underline{640 \text{ moves.}}}$$

Note :-

if head position is not given, take it as 0.

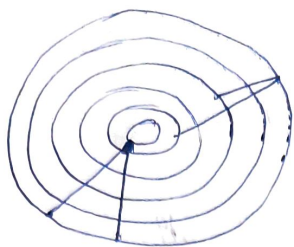
(2) SSTF (Shortest seek Time First)

Initial head position = 53

$$\begin{aligned} &|53 - 65| + |65 - 67| + |67 - 37| + |37 - 14| + |14 - 98| + \\ &|98 - 122| + |122 - 124| + |124 - 183| \end{aligned}$$

$$\Rightarrow 12 + 2 + 30 + 23 + 84 + 24 + 2 + 59 \Rightarrow 236 \text{ // } \text{movements}$$

(3) SCAN (Elevator Algorithm)



It will move till 0.

Given,

Initial head position = 53

Request sequence = 98, 183, 37, 122, 14, 124, 65, 67

Start from 53 and move towards right

Then it serves, 65, 67, 98, 122, 124, 183

Now, ~~Reverse~~ the direction and move towards left : 37, 14

$$\begin{aligned} \text{Total Track movement} &= |53 - 65| + |65 - 67| \\ &+ |67 - 98| + |98 - 122| + |122 - 124| + |124 - 183| \\ &+ |183 - 37| + |37 - 14| + |14 - 0| \end{aligned}$$

$$\Rightarrow 12 + 2 + 31 + 24 + 2 + 59 + 146 + 23 + 14$$

$$\Rightarrow \underline{\underline{313 \text{ moves.}}}$$

(4.) C-SCAN (Circular scan)

Given,

Initial head position = 53

Request sequence = 98, 183, 37, 122, 14, 124,
65, 67

Start from 53, move towards Right and

it serves, 65, 67, 98, 122, 124, 183. When
reaches the end, it jumps to the beginning
of the disc and now it serves, 14, 37.

$$\begin{aligned}\text{Total head movement} &= |53 - 65| + |65 - 67| \\ &+ |67 - 98| + |98 - 122| + |122 - 124| + |124 - 183| \\ &+ |183 - 0| + |0 - 14| + |14 - 37|\end{aligned}$$

$$\Rightarrow 12 + 2 + 31 + 24 + 2 + 59 + 14 + 23$$

$$\Rightarrow \underline{\underline{350 \text{ moves.}}}$$

(1.) Consider a logical address space of 64 pages with 1024 Bytes per page mapped to a physical memory of 256 frames.

Calculate :-

(a.) number of bits required for logical address

(b.) number of bits required for physical address.

Ans.) Given,

logical address space = 64 pages

page size = 1024 bytes per page

physical Memory space = 256 frames

(a) logical address = page number + offset

→ Bits required for page :

$$\text{no. of pages} = 64 = 2^6$$

So, 6 bits are needed, for the page numbers

→ Bits for offset :

$$\text{page size} = 1024 \text{ bytes} = 2^{10}$$

So, 10 bits are needed for the offset.

$$\text{Total bits for logical address} = 6 + 10 = \underline{\underline{16 \text{ bits}}}$$

(b) physical address = frame number + offset.

→ Bits for frame no :

$$\text{no. of frames} = 256 = 2^8$$

So, 8 bits are needed for the frame number

→ Offset Bits :-

offset is same as page size.

$$\text{offset} = \text{page size} = 1024 = 2^{10}$$

So, Total bits for physical address

$$\Rightarrow 8 + 10 = \underline{\underline{18 \text{ Bits}}}$$

(2)

Consider a logical address A of ~~pages~~ with a logical size of 512 bytes mapped in a logical memory of 64 bits. Find the no. of bits in the logical & physical address?

Given,

$$\text{Page no.} = 256 = 2^8$$

$$\text{Page size / offset} = \cancel{64 \text{ bits}} \quad 512 \text{ bytes} \\ \Rightarrow 2^9$$

$$\text{Frame size} \Rightarrow 64 \text{ bits} \Rightarrow \underline{\underline{2^6}}$$

$$\begin{aligned} * \quad \underline{\text{Physical Address}} &= \text{frame size} + \text{offset} \\ &\Rightarrow 6 + 9 = \underline{\underline{15 \text{ bits}}} \end{aligned}$$

$$\begin{aligned} * \quad \underline{\text{Logical Address}} &\Rightarrow \text{page} \cancel{\text{size}} \text{ no} + \text{offset} \\ &\Rightarrow 8 + 9 = \underline{\underline{17 \text{ bits}}} \end{aligned}$$